Differential Gene Expression and Overrepresentation Analysis of *Toxoplasma gondii* Infection in *Mus musculus*

RNA sequencing course

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Abstract

Toxoplasma gondii is a widespread parasite that induces strong immune responses in infected hosts. This project aimed to characterize host transcriptional changes associated with *T. gondii* infection using RNA sequencing data from mouse lung tissue. RNA-seq reads from infected and uninfected wildtype (WT) mice, as well as interferon α/γ receptor double knockout (DKO) mice, were analyzed using a standardized workflow including quality control, read alignment, gene quantification, differential expression analysis, and analysis of overrepresentation of Gene Ontology terms.

The results revealed extensive transcriptional remodeling following infection in both genotypes, with a predominance of down-regulated genes alongside strong induction of immune-related transcripts. Genotype-based comparisons showed a markedly stronger transcriptional response in WT mice compared to DKO mice, underscoring the importance of intact interferon signaling. Key interferon-responsive genes, including *Irf7* and *Rtp4*, were among the most significantly differentially expressed genes. Overrepresentation analysis further identified enrichment of immune-related biological processes in WT mice.

Together, these findings highlight the central role of interferon signaling in host responses to *T. gondii* infection and provide a foundation for future functional investigations.

Contents

1. Introduction

Differential gene expression (DGE) analysis is a powerful tool used to identify transcriptomic shifts in response to immune challenges. It brings crucial insights into disease mechanisms and immune regulation [1]. This project investigates RNA expression changes in mice infected with *Toxoplasma gondii*, a globally prevalent parasite. It's infection is estimated to one-third of the human population [2], [3]. While infection is often asymptomatic in immunocompetent individuals, *T. gondii* poses severe risks to immunocompromised patients, where it can cause encephalitis. In addition, primary infection during pregnancy can cause congenital toxoplasmosis, resulting in severe neurological damage or even fetal death [2], [4].

Exploring DGE is essential to decode the complex interaction between host and pathogen, where transcriptomic changes reflect both protective immune responses and parasite-driven modulation. Each insight contributes to the long-term goal of improving therapeutic targets and vaccine development [4].

This project is closely related to the study by Singhania et al. (2019), which investigated transcriptional immune responses to *Toxoplasma gondii* infection in mice. Their work demonstrated that interferon (IFN)-inducible transcriptional signatures are strikingly well preserved between lung tissue and blood, highlighting the extent to which immune pathways are shared between tissues versus remaining tissue-localized [1].

Their work showed that IFN-γ is the dominant cytokine required for survival during acute infection and for maintaining chronic control. Notably, many genes classically associated with Type I interferon signaling were found to be induced in an IFN-γ–dependent manner, highlighting extensive crosstalk between interferon pathways. To dissect the contribution of interferon signaling to these responses, mice lacking both type I and type II interferon receptors, thereby abolishing canonical interferon-mediated transcriptional responses, were part of the project [1], [4], [5].

High-throughput RNA sequencing is required to capture these responses due to its large dynamic range and sensitivity. The workflow of this project included quality control with FastQC, read alignment using Hisat2, gene quantification with featureCounts, differential expression analysis with DESeq2, and overrepresentation analysis of Gene Ontology terms using clusterProfiler. Together, these analyses enable the identification of infection- and genotype-associated transcriptional changes and provide insight into the biological processes underlying host responses to *T. gondii* infection.

The present project applies a standardized RNA-seq workflow to identify differential gene expression between infected and uninfected mice, as well as between wildtype and interferon α/γ receptor double knockout backgrounds. The analysis contextualizes these transcriptional changes within the framework established by Singhania et al. (2019) by assessing whether interferon-responsive immune signatures can be detected in the selected tissue using differential expression and functional enrichment analyses.

2. Methods

2.1 Preparation of repository

To initialize the project, a git repository was created locally, the cluster repository was connected to git. All scripts, containers, as well as some results can be found on GitHub [6], commits were done carefully and after each progression. Analyses were performed using containerized software executed via Apptainer on a high-performance computing cluster. Additional details on the sequencing data are available as "data_README.md" and "reads_README.md" in the GitHub repository. In this report, "Case" denotes *Toxoplasma gondii* -infected mice and "Control" denotes uninfected mice.

2.2 Quality Control

To assess the quality of the provided reads, FastQC (v0.12.1) was run on the lung reads. Detailed information about the preparation of the reads can be found in the GitHub repository in the notes folder. FastQC is a Java scripted quality control tool for high throughput sequence data [7]. The FastQC container image was loaded using Apptainer. FastQC results in .html files that were visualized in VS code and quality was controlled. In addition to that, the number of reads per sample was determined with a grep command (one "@" per read of 76 bp), and forward and reverse samples were compared.

2.3 Mapping reads to reference genome

For mapping the next-generation sequencing RNA reads Hisat2 (v2.2.1) was used, a sensitive and fast alignment program which can map reads to a reference genome. In our case, the latest reference genome sequence from *Mus musculus* (GRCm39) was downloaded from the Ensembl ftp site (release 115) [8] and index files required for Hisat2 were generated with strandedness setting RF. Execution results in separate SAM files for each sample, which were further converted to BAM format and sorted by genomic coordinates using Samtools (v1.20). Samtools is a set of utilities that converts, indexes, merges and sorts between the formats SAM, BAM and CRAM [9].

2.4 FeatureCounts - counting numbers of reads per gene

Gene-level quantification was performed using featureCounts from the Subread package (v2.0.6), executed within an Apptainer container. A table of counts containing the number of reads per gene in each sample was generated. The program counts mapped reads for features such as genes, takes SAM or BAM files as well as an annotation file as input, and outputs the numbers of reads assigned to the wanted feature. A result summary is generated that includes number of successfully assigned as well as failed assigned reads [10].

2.5 Exploratory data and differential expression analysis

For the exploratory data analysis the table of counts from the previous step was used as starting material. All further steps were done locally with R in RStudio (R version 4.5.2 (2025-10-31), RStudio version 2025.09.2). For the exploratory and differential expression analysis DESeq2 (v1.50.2) was used. DESeq2 is a Bioconductor 3.22 software package that tests count data from high-throughput sequencing assays for differential expression (DE) by using a negative bionomial distribution model [11]. The table was reformatted to match the expectation of DESeq2 and the DESeqDataSet object was created. Then, dependence of the variance on the mean was removed using vst and sample clustering was visualized with a PCA plot.

Further, results from the DE analysis were extracted with three different pairwise contrasts. First, both wild type groups (WT Case, WT Control), both double knock-out groups (DKO Case, DKO Control) and lastly both WT and DKO Cases were used as pairwise contrast. For each comparison, the reference level of the DESeq2 design was adjusted accordingly to ensure that the extracted contrasts matched the intended experimental conditions. For visualization of the DE results, both heatmaps and volcano plots were generated using the packages pheatmap (v1.0.13) and EnhancedVolcano (v1.28.2). Heatmaps were constructed using the 50 genes with the lowest adjusted p-values for each contrast, focusing on the changes in gene expression with the most statistical significance. Variance stabilized transformed (vst) expression values were used.

To facilitate biological interpretation and visualization of the results, Ensembl gene identifiers (ENSMUSG IDs) were mapped to official mouse gene symbols (MGI symbols). This was performed using the biomaRt package (v2.66.0), accessing the Ensembl database with the functions useMart() and getBM() to retrieve the corresponding gene symbols for each Ensembl gene ID. For the generation of volcano plots, genes were ranked by multiplying the absolute $\log_2$ fold change with the negative logarithm of the adjusted p-value for each gene. This ranking prioritizes genes that exhibit high expression change and high statistical significance, which are both significant properties for volcano plots. The genes with the ten highest ranks were associated with their points in the corresponding plot.

2.6 Overrepresentation analysis

Overrepresentation analysis was performed using the Bioconductor package clusterProfiler (v4.18.2), which provides statistical methods for functional interpretation of omics data. Gene Ontology (GO) enrichment was conducted using the enrichGO() function, specifying the Biological Process (BP) ontology and ENSEMBL gene identifiers as the key type. Genome-wide mouse annotation was supplied via the org.Mm.eg.db package (v3.22.0) as the OrgDb reference [12]. Barplots were restricted to the top ten enriched GO categories to provide a concise and interpretable overview of the overrepresentation analysis results.

3. Results

3.1 Quality Control

Overall read quality assessed by FastQC was sufficient for downstream analysis. Elevated sequence duplication levels and deviations in per-base sequence content were observed in some samples, consistent with highly expressed transcripts and library preparation effects commonly seen in RNA-seq data [7]. No adapter contamination was detected. A mild deviation in per-sequence GC content was observed for forward reads only, consistent with strand-specific protocols, and as no quality issues were identified, no read trimming was performed. Read counts ranged from approximately 30,000 to 55,000 per sample (see Table 1), and all samples were retained for downstream analysis.

3.2 Mapping with Hisat2

Mapping of reads to the mouse reference genome was performed using Hisat2, followed by conversion of SAM files to BAM format, sorting, and indexing. All steps were completed successfully for every sample. Alignment summary statistics indicated consistently high mapping quality across samples. All libraries showed 100% paired-end reads, and the overall alignment rate did not drop below 94.14% for any sample (Table 1), indicating efficient and reliable read mapping. Concordant alignment refers to paired-end reads that align with the expected orientation and within the expected insert size range [13]. Across all samples, a high proportion of read pairs aligned concordantly exactly once, with values consistently above 86.75%. This suggests that the majority of reads mapped uniquely and in a biologically plausible manner, supporting the suitability of the data for downstream quantitative analyses.

Alignment statistics indicated high-quality mapping, with an overall alignment rate of above 94% for all samples. Over 88% of read pairs aligned concordantly, the majority mapping uniquely. A small proportion of reads mapped to multiple genomic locations, which is expected in RNA-seq data due to shared sequence features among transcripts and gene families. These multimapped reads do not pose a concern for downstream differential expression analysis [14].

3.3 FeatureCounts - counting numbers of reads per gene

Analysis with FeatureCounts confirmed a high proportion of assigned reads, with 77% or more assigned in all samples and a mean assignment rate of 79.88%. Consequently, approximately 20.12% of reads remained unassigned. These unassigned reads could reflect a combination of ambiguous overlaps between multiple genes, reads mapping to intergenic regions and multimapped reads, which are common in RNA-seq experiments. In general, reads cannot be assigned unambiguously when they overlap multiple genes, originate from highly similar gene families, or map to regions shared between transcript isoforms [15], [16].

3.4 Exploratory analysis

To visualize sample clustering based on global gene expression profiles, a principal component analysis (PCA) was performed using rlog-transformed counts (Figure 1). In this

Sample	Group	# r.p.s.	# as. r.p.s.	as. r.p.s. [%]	al. rate [%]
18	Lung_WT_Case	36220112	29665762	81.90	97.32
19	Lung_WT_Case	36285942	29275580	80.68	97.37
20	Lung_WT_Case	39982873	29659751	74.18	94.14
21	Lung_WT_Case	40592768	32653603	80.44	97.28
22	Lung_WT_Case	37737087	30209735	80.05	97.34
23	Lung_DKO_Case	38564395	30670071	79.53	96.02
24	Lung_DKO_Case	54744146	43442785	79.36	96.96
25	Lung_DKO_Case	34754674	26983807	77.64	96.19
27	Lung_DKO_Case	31608859	24370564	77.10	96.68
37	Lung_WT_Control	44881449	36370705	81.04	97.52
38	Lung_WT_Control	40988016	34801956	84.91	97.25
39	Lung_WT_Control	46697727	37759512	80.86	98.09
40	Lung_DKO_Control	36802873	28960107	78.69	96.38
41	Lung_DKO_Control	39406044	32195278	81.70	97.65
42	Lung_DKO_Control	44301194	35514137	80.17	97.45

Table 1: Overview of RNA-seq samples (original sample name is SRR78219 + the number in the table) with group assignment, number of reads per sample, number of assigned reads per sample after mapping, percentual number of assigned reads per sample and overall alignment rate in percent. All values were the result of FeatureCounts analysis.

Comparison	DE genes (padj < 0.05)	Up-reg.	Down-reg.
WT Case vs WT Control	10815	5043	5772
DKO Case vs DKO Control	11226	5470	5756
WT Case vs DKO Case	7805	4232	3573

Table 2: Differential expression summary statistics for the three contrasts.

representation, each point corresponds to one sample, and the first two principal components (PC1 and PC2) capture the largest sources of variance in the dataset. The percentages on the axes reflect the proportion of total variance explained by the respective component [17].

In the PCA plot, samples generally clustered according to their experimental group. Replicates of each group are positioned close together, although the lung WT control group displays greater dispersion compared to the WT infected group. Separation between infected and control samples is visible, as well as between wildtype and interferon receptor double knockout (DKO) samples.

3.5 DE analysis

Table 2 summarizes the number of differentially expressed genes (DEGs) identified with DESeq2 using an adjusted p-value threshold of < 0.05. In both infection-based comparisons (WT Case vs WT Control and DKO Case vs DKO Control), the number of down-regulated genes is bigger compared to the number of up-regulated genes. In contrast, the genotype-based comparison (WT Case vs DKO Case) shows a higher proportion of up-regulated genes.

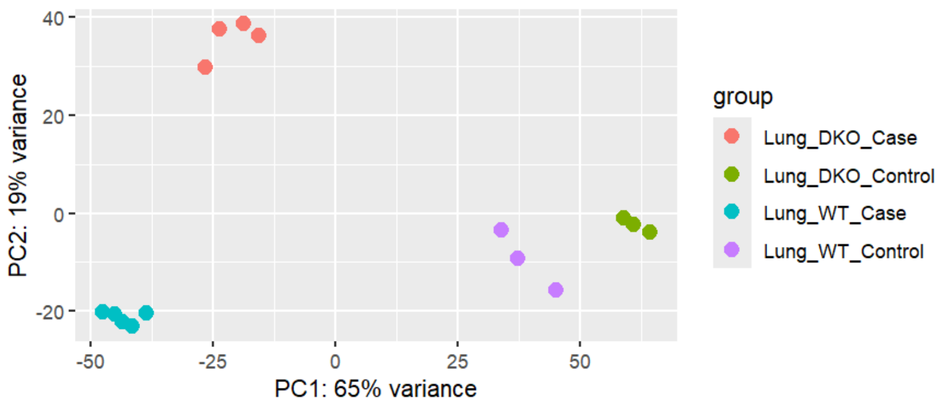

Figure 1: PCA plot with rlog-transformed gene expression data of the four different sample categories. PC1 and PC2 show the largest proportions of variance.

Volcano plots were generated for each pairwise comparison to visualize the magnitude and significance of differential expression (Figure 2). Each plot displays all genes with the ten most significant DEGs annotated. In the WT Case versus DKO Case comparison, several adjusted p-values were extremely small and treated as zero in R. To generate the plot, these values were adjusted algorithmically to maintain an ordered ranking relative to the next lowest value, resulting in reduced dispersion of data points in this panel.

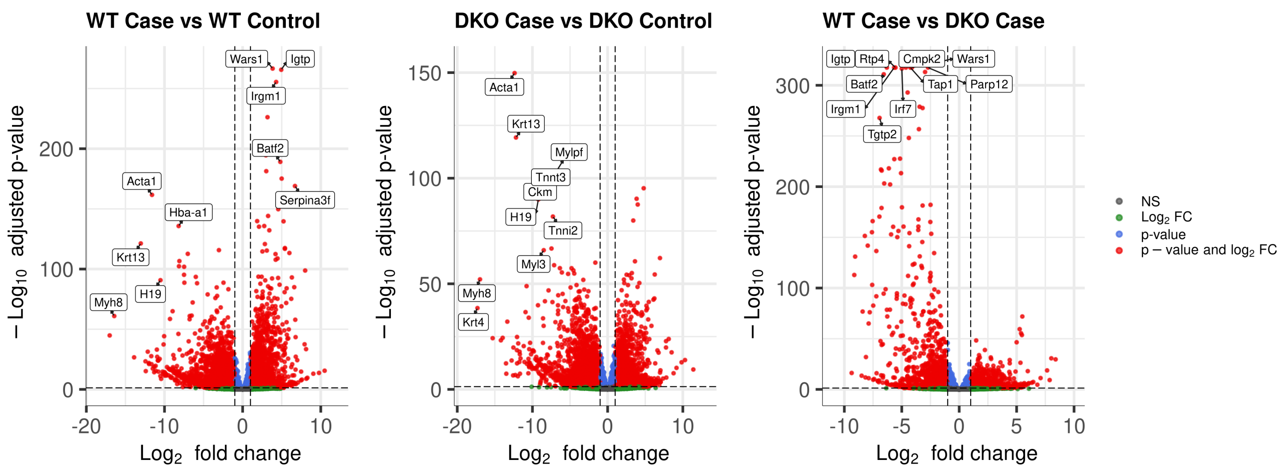

Figure 2: Volcano plots of the log 2 fold change plotted against the adjusted p-value for all three comparisons. Top 10 genes are assigned to their corresponding point in the plot. Calculation of ranking was done as explained in method section 2.5

3.6 Overrepresentation analysis

Overrepresentation analysis was performed using clusterProfiler, and the results are displayed as barplots showing the ten GO terms with the lowest p-values for each comparison. In the WT Case versus Control comparison, the top three enriched biological processes were *muscle system process*, *muscle contraction*, and *taxis*. In the DKO Case versus Control comparison, the three most enriched processes were *muscle system process*, *muscle contraction*, and *muscle cell differentiation*. For the WT Case versus DKO Case comparison, the highest-ranking enriched processes were *positive regulation of response to biotic stimulus*, *leukocyte migration*, and *positive regulation of innate immune response*.

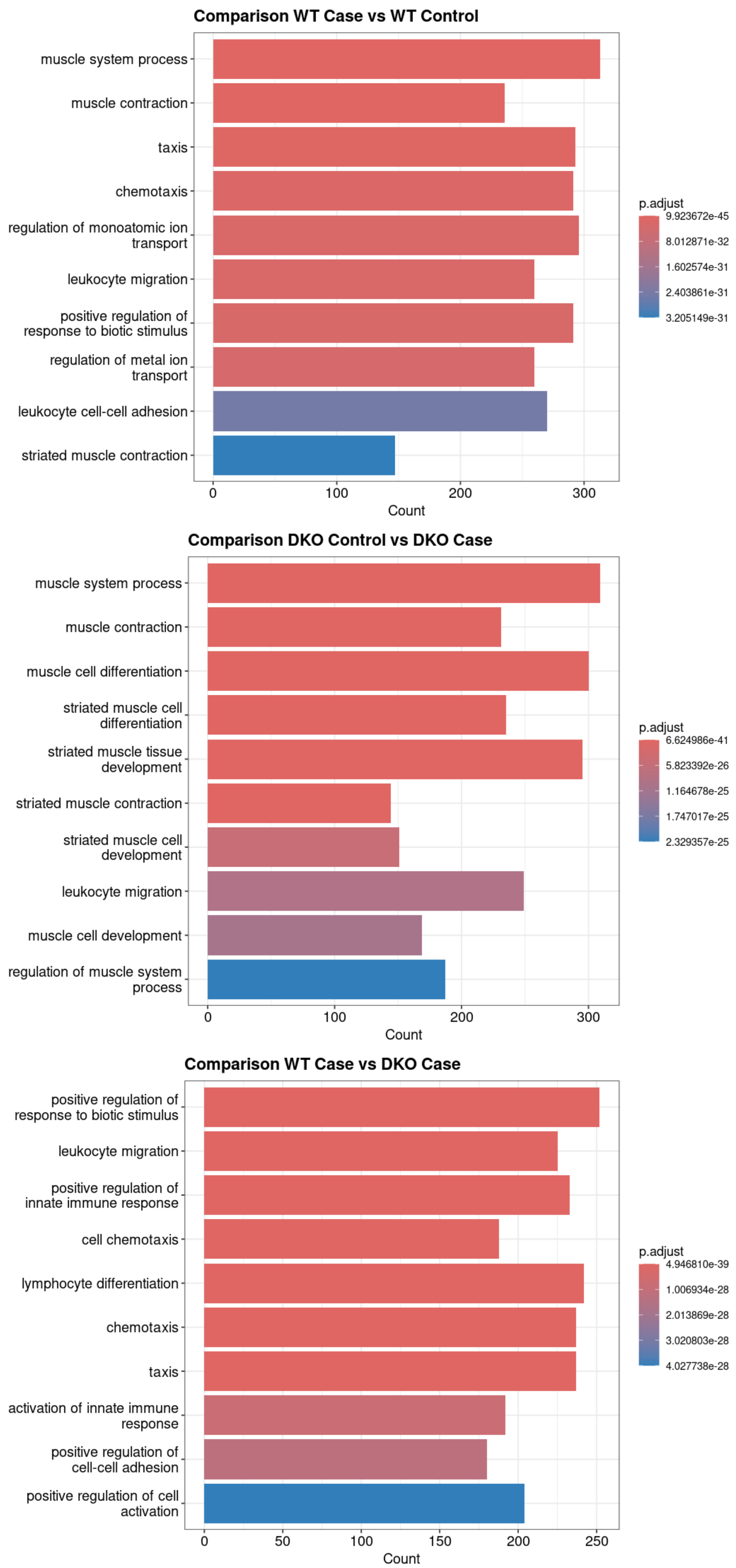

Figure 3: Overrepresentation analysis for the top 10 GO terms in each pairwise comparison. Each panel shows the 10 most enriched categories.

4. Discussion

QC and mapping statistics confirmed that sequencing reads were of sufficient quality and mapping efficiency, supporting the validity of downstream differential expression and enrichment analyses.

4.1 Exploratory analysis

The PCA results indicate that both infection status and host genotype alter transcriptomic profiles. Clustering of biological replicates demonstrates good consistency within experimental groups, while the greater dispersion observed in the lung WT control samples could reflect natural biological variability in uninfected tissue. Clear separation of case and control samples suggests that *T. gondii* infection strongly alters gene expression patterns. Additionally, differences between wildtype and DKO samples imply a genotype-dependent effect on transcriptional responses, matching the expected role of interferon signaling in shaping host immune response.

4.2 Differential expression analysis

Differential expression analysis revealed distinct transcriptional responses depending on infection status and genotype. In both infection-based comparisons, a greater number of down-regulated than up-regulated genes suggests a net reduction in global gene expression following infection, despite strong induction of specific immune-related genes. This points to targeted immune activation accompanied by suppression of housekeeping or metabolic processes. In contrast, the genotype-based comparison between WT Case and DKO Case showed a higher proportion of up-regulated genes in WT mice, reflecting a more robust transcriptional response when interferon α/γ receptor signaling is intact. In DKO mice, the absence of this signaling impairs efficient induction of immune-response genes, resulting in attenuated transcriptional activation [18].

Volcano plot analysis highlighted key DEGs with large fold changes and high statistical significance. Comparison with Singhania et al. (2019) identified *Irf7* and *Rtp4* among prominent genes in the WT Case versus DKO Case comparison. Both genes are interferon-responsive, and their DE aligns with the loss of interferon receptor signaling in DKO mice [19]. Although *RTP4* is best known as a chaperone for bitter taste receptors, it is also a recognized interferon-stimulated gene. Its regulation in this dataset likely reflects interferon-driven immune activation during *T. gondii* infection, consistent with previous reports [20, 21].

4.3 Overrepresentation analysis

Overrepresentation analysis revealed distinct biological processes associated with infection and genotype. In both infection-based comparisons, enrichment of muscle contraction–related terms might reflect infection-induced changes in lung tissue structure or cellular composition. Enrichment of taxis- and differentiation-related terms suggests different cell migration and tissue remodeling during infection.

By contrast, the WT Case versus DKO Case comparison showed strong enrichment of immune-related processes, including leukocyte migration and regulation of innate immune responses. These findings support requirement for intact interferon signaling to induce effective immune-response pathways.

4.4 Conclusion

Overall, this study demonstrates that *T. gondii* infection induces substantial transcriptional remodeling in the lung, with interferon signaling playing a central role in shaping genotype-dependent immune responses. A more extensive analysis could explore the functional roles of the identified differentially expressed genes in more detail. The role of DE genes like *Igtp* or *Irgm1* in the IFN-γ pathway could be further investigated, providing deeper insight into the molecular mechanisms underlying the host response to infection.

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5. Disclosure of use of AI-assisted technologies

During the preparation of this work, I used the following tools:

1. https://chat.openai.com/ - for grammar, vocabulary and sentence structure as well as for generating, optimizing and troubleshooting python code.

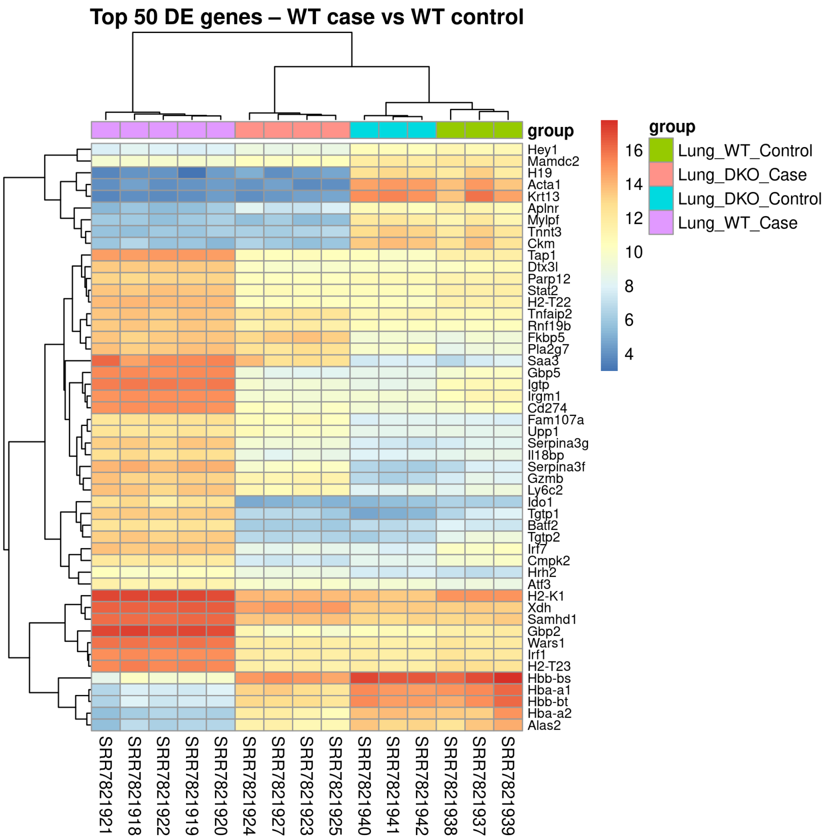
2. https://notebooklm.google.com/ - for finding sources related to the topic and finding information within the sources.

2. https://www.leo.org/englisch-deutsch - for vocabulary and word translation.

After using these tools/services, I reviewed and edited the content as needed and take full responsibility for the content of the publication. I am aware that in case of dis-compliance, the generated text is considered plagiarism with its legal consequences.

6. Appendix

6.1 Heatmaps



(a) WT Case vs WT Control

Figure 4: Heatmaps of the top 50 differentially expressed genes for the indicated pairwise comparisons. Expression values are based on variance-stabilized counts.

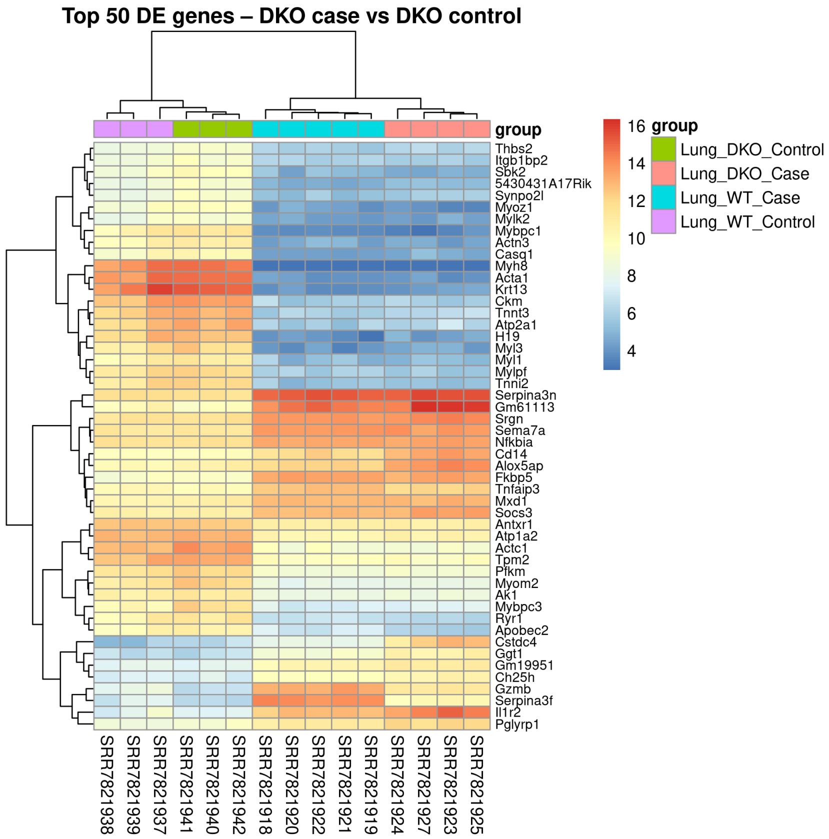


Figure 5: Heatmaps of the top 50 differentially expressed genes for the indicated pairwise comparisons (continued).

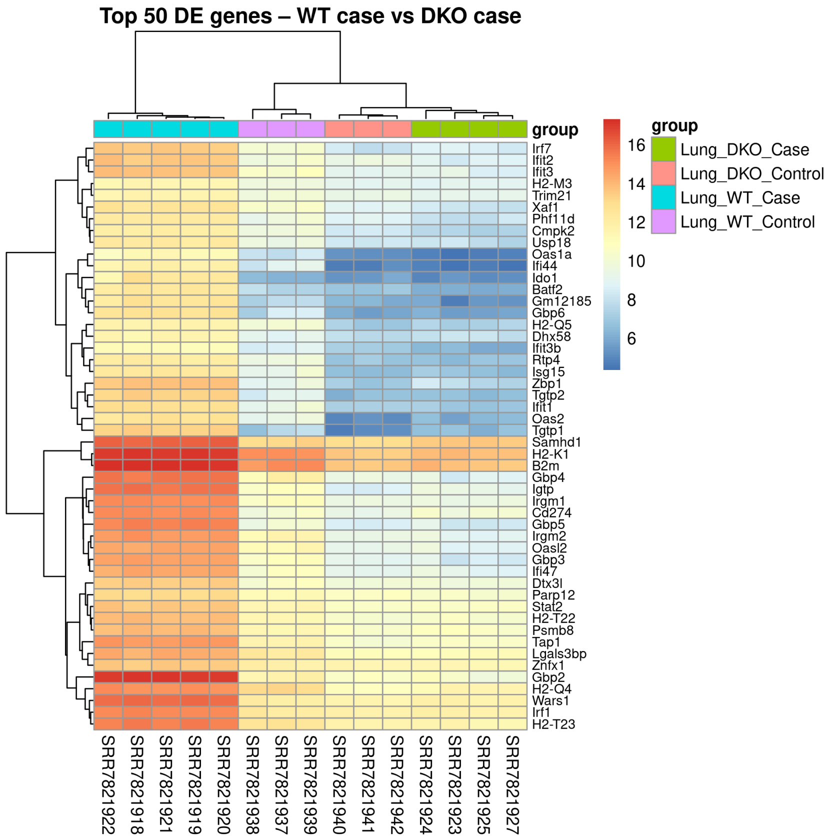


Figure 6: Heatmaps of the top 50 differentially expressed genes for the indicated pairwise comparisons (continued).